Assignment Report

Homology Modeling

Group Number- 11

Group members-

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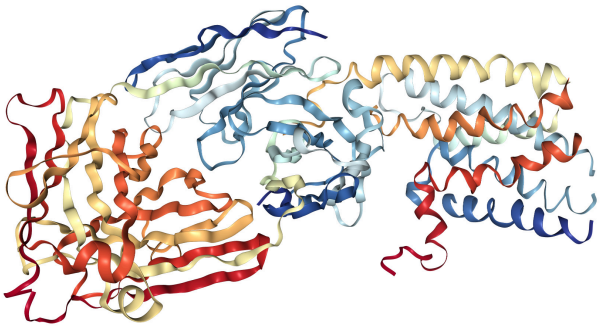
- Bhavya
- Vaibhav Samrat Waghaye
- Shlok Gupta
- Ammar Jamil

Protein Assigned-

- G protein-coupled receptor (β2-adrenergic)

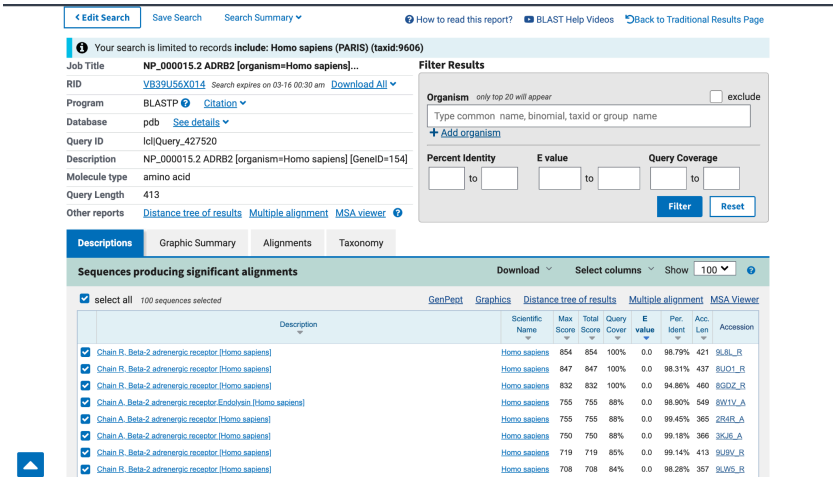
UniProt / NCBI ID-

- P07550 (ADRB2_HUMAN)(Uniprot ID)



The β2 adrenergic receptor (ADBR2) is a G protein-coupled receptor that plays a key role in the fight-or-flight response, via binding epinephrine/adrenaline. It is a seven-transmembrane protein that activates the Stimulatory G-protein, stimulating adenylate cyclase to increase cyclic AMP. Primarily, β2 receptors relax smooth muscle, particularly in airways.

Section 1: Template Identification

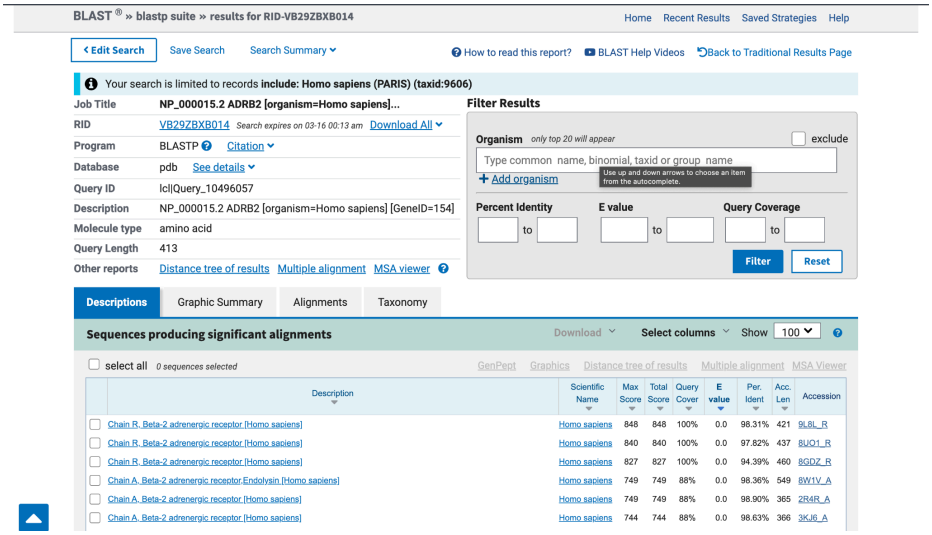
Blast Result Page Snapshot	
Original FASTA sequence	>NP_000015.2 ADRB2 [organism=Homo sapiens] [GeneID=154] MGQPGNGSAFLLAPNGSHAPDHDVTQERDEVWVVGMGIVMSLI VLAIVFGNVLVITAIAKFERLQTVTNY FITSLACADLVMGLAVVPFGAAHILMKMWTFGNFWCEFWTSIDV LCVTASIETLCVIAVDRYFAITSPFK YQSLLTKNKARVIILMVWIVSGLTSFLPIQMHWYRATHQEAINCYA NETCCDFFTNQAYAIASSIVSFYV PLVIMVFVYSRVFQEAKRQLQKIDKSEGRFHVQNLSQVEQDGRT GHGLRRSSKFCLKEHKALKTLGIIMG TFTLCWL**P**FFIVNIVHVIQDNLIRKEVYILLNWIGYVNSGFN**P**LIY CRSPDFRIAFQELLCLRRSSLKAY GNGYSSNGNTGEQSGYHVEQEKENKLLCEDLPGTEDFVGHQG TVPSDNIDSQGRNCSTNDSLL

Blast Record Table (Wild-Type) -

S.No.	PDB ID	Sequence Identity (%)	Query Coverage (%)	Resolution (Å)	Organism
1.	**9U9V (Chain R)**	99.14%	85%	2.85	*Homo sapiens*
2.	**3KJ6 (Chain A)**	99.18%	88%	2.85	*Homo sapiens*
3.	**9L8L (Chain R)**	98.79%	100%	3.22	*Homo sapiens*

Section 2: Modeling the Wild-Type and Mutants

Part A: Mutation Introduction

Blast Result Page Snapshot (Mutated)	
Mutated FASTA Sequence	>NP_000015.2 ADRB2 [organism=Homo sapiens] [GeneID=154] MGQPGNGSAFLLAPNGSHAPDHDVTQERDEVWVVGMGIVMSLIVLAIVFGN VLVITAIAKFERLQTVTNY FITSLACADLVMGLAVVPFGAAHILMKMWTFGNFWCEFWTSIDVLCVTASIET LCVIAVDRYFAITSPFK YQSLLTKNKARVIILMVWIVSGLTSFLPIQMHWYRATHQEAINCYANETCCDF FTNQAYAIASSIVSFYV PLVIMVFVYSRVFQEAKRQLQKIDKSEGRFHVQNLSQVEQDGRTGHGLRRS SKFCLKEHKALKTLGIIMG TFTLCWL**S**FFIVNIVHVIQDNLIRKEVYILLNWIGYVNSGFN**T**LIYCRSPDFRI AFQELLCLRRSSLKAY GNGYSSNGNTGEQSGYHVEQEKENKLLCEDLPGTEDFVGHQGTVPSDNID SQGRNCSTNDSLL (pink color demonstrates- mutations)

Residue Table-

S.No.	Residue Number	Original Residue	Mutated Residue	Remark
1.	288	PRO (P)	SER (S)	Non-polar → Polar. Part of the CWxP motif; disrupts the functional helix kink.
2.	323	PRO (P)	THR (T)	Non-polar → Polar. Part of the NPxxY motif; affects structural conservation.

Snapshots with labels provided in Section 3 (Chimera Visualization).

Blast Record Table (Mutants) -

S.No.	**PDB ID**	**Sequence Identity (%)**	**Query Coverage (%)**	**Resolution (Å)**	**Organism**
1.	**9U9V (Chain R)**	**98.57%**	**85%**	**2.85**	*Homo sapiens*
2.	**3KJ6 (Chain A)**	**98.63%**	**88%**	**2.85**	*Homo sapiens*
3.	**9L8L (Chain R)**	**98.31%**	**100%**	**3.22**	*Homo sapiens*

Part B: Homology Modeling

WT Model ID	MOLPDF Score	DOPE Score	Model evaluation
PROTEIN.B99990001.pdb	12775.94922	-44382.027344	Lowest MOLPDF
PROTEIN.B99990002.pdb	12949.49707	-44152.312500	Worst MOLPDF and DOPE
PROTEIN.B99990003.pdb	12891.10742	-45060.601563	Best DOPE

MT Model ID	MOLPDF Score	DOPE Score	Model evaluation
MUTANT.B99990001.pdb	12771.36426	-44906.792969	Lowest MOLPDF
MUTANT.B99990002.pdb	12788.31152	-44924.753906	Best DOPE
MUTANT.B99990003.pdb	12855.30762	-44576.769531	Worst DOPE and MOLPDF

Model quality criteria-

- MOLPDF: Lower value indicates better optimized model
- DOPE: More negative value indicates better structural quality and stability
- Better DOPE score given preference over better MOLPDF

We did model evaluation using MOLPDF and DOPE scores.

1) Wild Type Models-
    - PROTEIN.B99990003.pdb had the best DOPE score (-45060.601563) which indicates the most stable and reliable structure
    - PROTEIN.B99990001.pdb had the lowest MOLPDF score which indicates good model optimization
    - PROTEIN.B99990002.pdb had the least favourable DOPE score and MOLPDF score which makes it the weakest model among the wild type structures

2) Mutant Type Models-
    - MUTANT.B99990002.pdb had the best DOPE score (-44924.753906) which indicates the better structural quality among the mutant models
    - MUTANT.B99990001.pdb had the lowest MOLPDF score which indicates good model optimization
    - MUTANT.B99990003.pdb had the least favourable DOPE score and MOLPDF score which makes it the weakest model among the mutant types

Final model selection - Based on DOPE preference , PROTEIN.B99990003.pdb (wild type) and MUTANT.B99990002.pdb (mutant) were selected as the best representative models.

File Snapshots-

Wild type-

```
>> Summary of successfully produced models:
Filename                          molpdf
----------------------------------------
WT9U9V.B99990001.pdb         12775.94922
WT9U9V.B99990002.pdb         12949.49707
WT9U9V.B99990003.pdb         12891.10742
```

```
<< end of ENERGY.
DOPE score               : -44382.027344
('DOPE SCORE:', -44382.027344)
```

```
<< end of ENERGY.
DOPE score               : -44152.312500
('DOPE SCORE:', -44152.312500)
```

```
<< end of ENERGY.
DOPE score               : -45060.601563
('DOPE SCORE:', -45060.6015625)
```

Mutant type-

```
>> Summary of successfully produced models:
Filename                          molpdf
----------------------------------------
MUTANT.B99990001.pdb         12771.36426
MUTANT.B99990002.pdb         12788.31152
MUTANT.B99990003.pdb         12855.30762

Total CPU time [seconds]                              :      71.17
```

```
<< end of ENERGY.
DOPE score               : -44906.792969
```

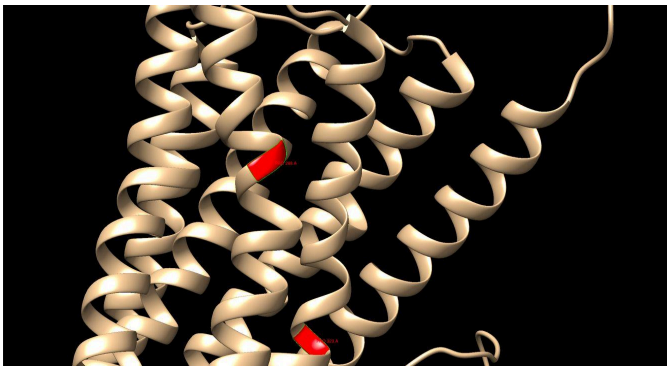
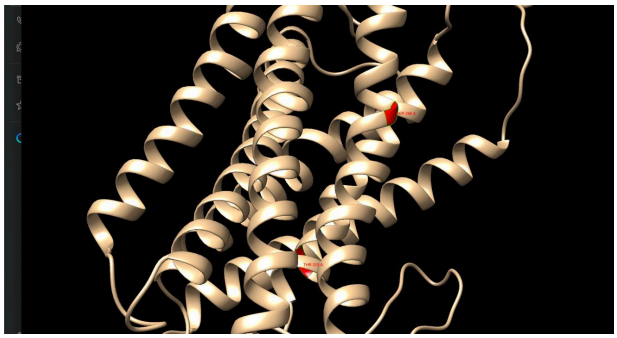
```
<< end of ENERGY.
DOPE score               : -44924.753906
```

```
<< end of ENERGY.
DOPE score               : -44576.769531
```

Section 3: Chimera Visualization

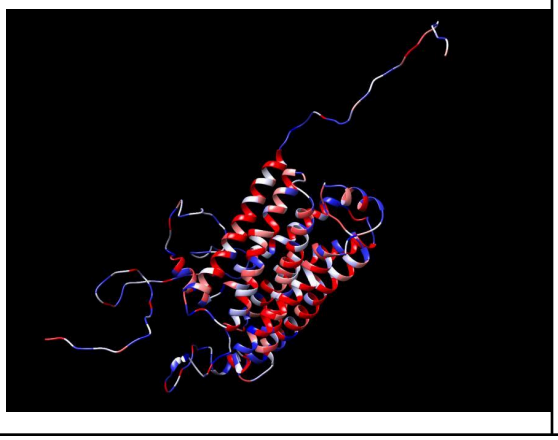
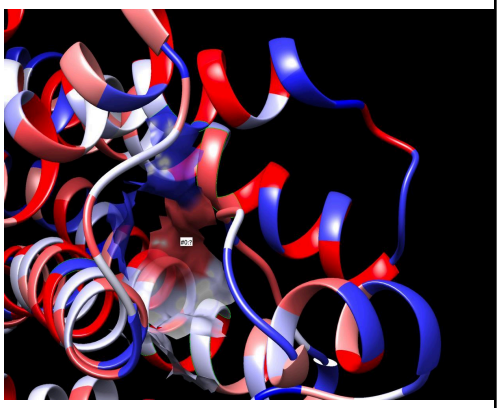
Snapshots with labels for Section 2, Part A (Mutation Introduction).

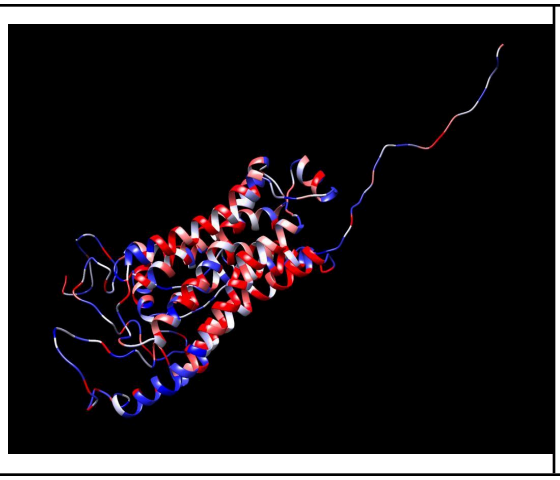
Highlighted residues within the Wild-type structures and the Mutant structures-

Wild-Type Residues (Before)	Mutant Residues (After)
	

Specific Chimera Tasks-

Color hydrophobic regions; Surface representation of binding pocket

	Hydrophobic Regions	Surface representation of binding pocket
Wild Type (Best Model)		

Mutant (Best Model)		

Section 4: Structural Comparison

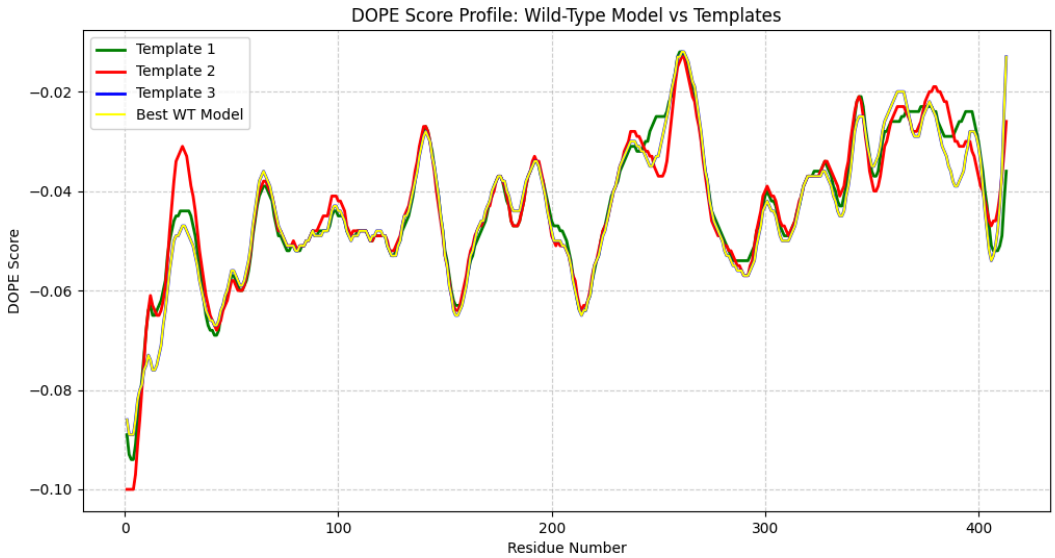
Part A: DOPE Score Analysis

From Section 2, Part B we determined-

- Best WT Model- PROTEIN.B99990003.pdb
- Best MT Model - MUTANT.B99990002.pdb

Dope Score Profile Plots Generated-

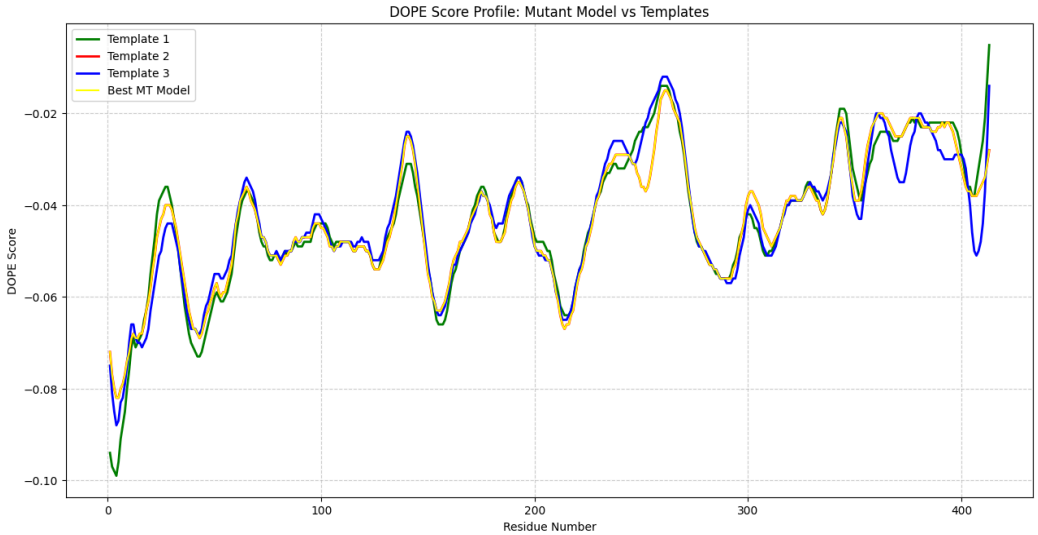
1. DOPE Score Profile – Wild-Type Model vs Templates



Note: Best WT model overlaps the line graph for template 3 as best wt model is template 3.

The DOPE score profile compares the per-residue energy of the best wild-type model with the three selected template structures. The WT model closely follows the energy trends of the templates across most residues, indicating that the homology modeling successfully captured the overall structural features of the protein. Regions with slightly higher (less negative) DOPE values correspond to flexible loop regions that are typically less constrained during modeling. Overall, the similarity between the WT model and templates suggests that the predicted structure is reliable and structurally consistent with the template proteins.

2. <u>DOPE Score Profile – Mutant Model vs Templates</u>



Note: Best MT model overlaps the line graph for template 2 as best mt model is template 2.

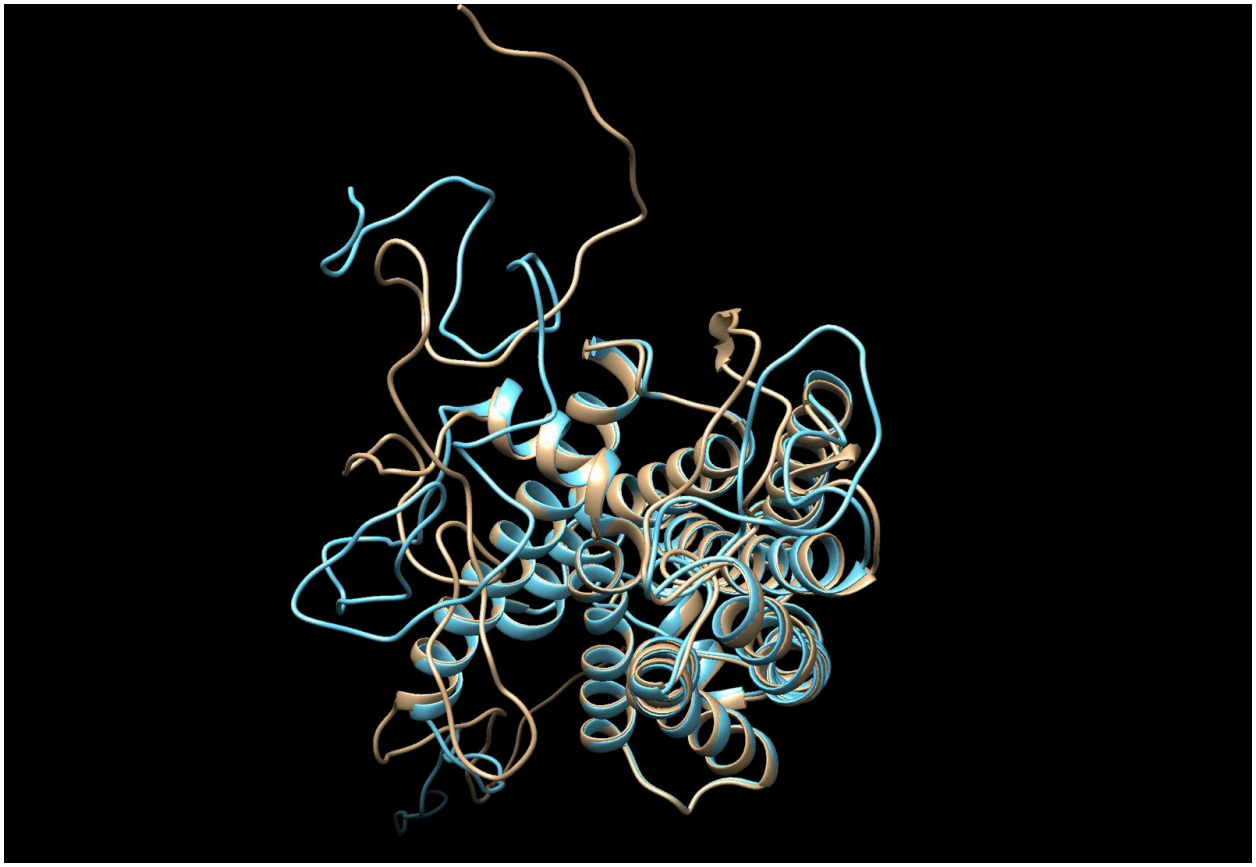
The DOPE score profile of the mutant model is compared with the same three template structures. The mutant model shows a similar energy pattern to the templates and the wild-type model across the sequence, indicating that the introduced mutations do not cause major global structural changes. Minor deviations in certain regions reflect localized energetic differences due to the residue substitutions. Overall, the mutant structure remains structurally stable and largely consistent with the template-derived fold.

Part B: RMSD Analysis

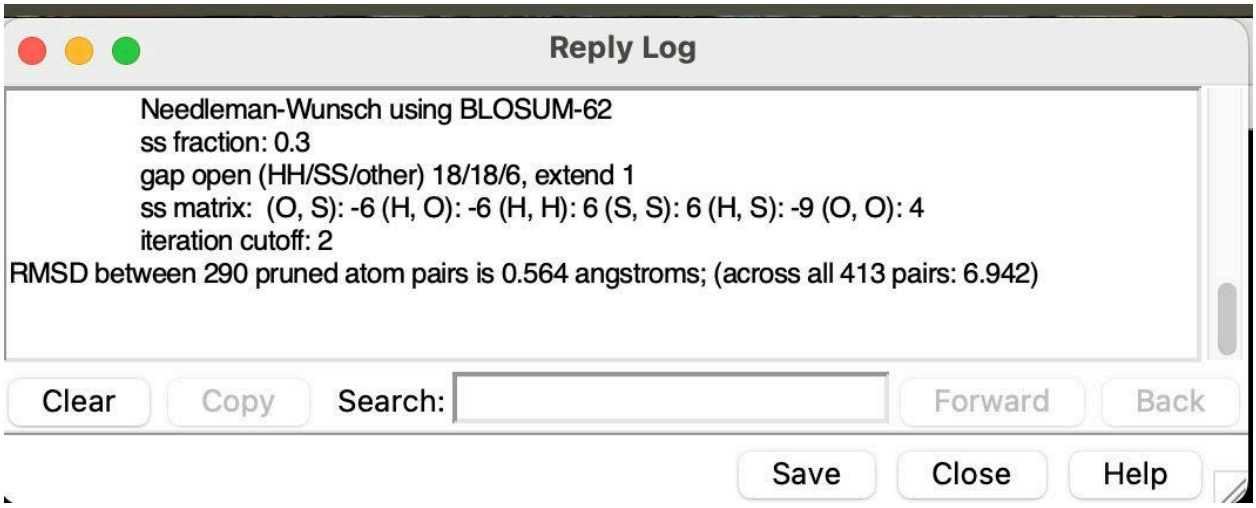
To compare the global structural divergence, the best mutant model was superimposed onto the best wild-type model using MatchMaker. The calculation resulted in an RMSD of 0.564 Å for 290 pruned atom pairs. This low RMSD value confirms that the overall seven-transmembrane architecture remains intact after the P288S and P323T mutations.

Superimposed Mutant type with wild type-

Gold: Best Wild-Type Model
Light Blue: Best Mutant Model



Reply log-



Part C: Interpretation & Insights

Using DOPE scores and RMSD values (relative to wild type protein) we evaluated the structural impact of mutations on the protein structure. These 2 metrics provide complementary information (i.e DOPE reflects the energy stability whereas RMSD quantifies structural deviation between the 2 structures)

1) **DOPE score-** We noticed that the wild type model had a lower DOPE score compared to the mutant structures , which means that it has a relatively more stable conformation. In contrast mutant models had higher DOPE scores which prove that substitutions reduce structural stability. This difference likely arises from the substitutions of proline residues with polar residues (Pro->Ser and Pro->Thr). Proline residues often play a key structural role in maintaining helix bends or motifs, replacing them with more flexible polar residues can disrupt local packing interactions and alter the stability of nearby structural elements.
2) **RMSD analysis-** After doing RMSD analysis(i.e superimposing the best mutant model and best wild type model and finding an RMSD value of approximately 0.564 Å) we noticed increased differences between the structures of wild type and mutant. Mutations which had small RMSD values show minor changes in local conformation whereas higher RMSD values show larger structural rearrangement which may affect the proteins folding or functional regions.

Hence in conclusion, the above results together show that mutations likely introduce some **destabilization** and changes in proteins conformation rather than a complete conformational change. These changes could influence the protein function by changing local interactions or structural flexibility of the protein. Such subtle structural changes can still be biologically significant as even small modifications in local structure may alter receptor activation or ligand binding.